Notebook naam nhanes_JOINT_S1_rfbalanced_k19_S2_rfbalanced_k12_CLEAN_FINA...

Ei notebook e
ei model er jonno ami eigula korte chai

Fine OBU search

Stage-2 calibration: sigmoid + isotonic

Stage-2 ensemble

Alternative Stage-2 models: LR, RF, ExtraTrees, HistGradientBoosting, XGBoost if available

Stage-2 cost-sensitive training

Error analysis + subgroup-specific threshold experiment

ei experiment gulo

resulted notebook nhanes_STAGE2_OBU_CALIBRATION_ENSEMBLE_ERROR_ANALYSI...

এই notebook-এ আছে:

1. **Fine OBU search**
   k_neighbors, majority_remove_frac, target_ratio grid search করে validation set থেকে best OBU setting select করবে।
2. **Stage-2 threshold fine tuning**
   Stage-2 threshold 0.25–0.70 range-এ fine search করবে।
3. **Stage-2 calibration**
   Baseline/best OBU Stage-2 model-এর উপর:
   - Sigmoid calibration
   - Isotonic calibration
4. **Alternative Stage-2 models**
   - Logistic Regression
   - Random Forest
   - ExtraTrees
   - HistGradientBoosting

- XGBoost, যদি environment-এ installed থাকে

5. **Stage-2 ensemble**
   Best Stage-2 candidates combine করে probability averaging ensemble বানাবে।
6. **Stage-2 cost-sensitive training**
   High-risk class weight বাড়িয়ে LR/RF/ExtraTrees/XGBoost variant test করবে।
7. **Error analysis**
   Best validation-selected candidate-এর:
   - confusion matrix
   - actual-to-predicted error decomposition
   - high→low miss
   - moderate→low miss
   - low→at-risk false alarm
   - subgroup-wise metrics
8. **Subgroup-specific threshold experiment**
   Validation set দিয়ে subgroup-specific Stage-2 threshold tune করবে, যেমন:
   - sex group
   - age group
   - BMI group
   - diabetes group
   - hypertension group
   - BP medicine group

Best useful improvement: Stage-2 threshold retuning

Stage-2 threshold retuning kore best threshold holo **0.50**.

Model	Accuracy	Balanced Acc	Macro F1	Moderate Recall	High Recall	High→Low Miss
Original baseline	0.7119	0.6451	0.5322	0.4476	0.7368	2
Re-thresholded baseline	**0.7191**	**0.6641**	**0.5591**	**0.5048**	**0.7368**	2

Fine OBU search ki korlo?

Validation set-e best OBU config chilo:

k_neighbors = 3
majority_remove_frac = 0.30
target_ratio = 1.25
stage2_threshold = 0.60

Validation result bhalo:

Metric	Validation
Accuracy	0.6987
Balanced Acc	0.6913
Macro F1	0.5548
Moderate Recall	0.6381
High Recall	0.7297

But test-e:

Metric	Test
Accuracy	0.7191
Balanced Acc	0.6417
Macro F1	0.5579
Moderate Recall	0.5429
High Recall	0.6316

Problem holo high recall **0.7368 theke 0.6316** e neme gese. Screening model-er jonno eta risky.

So **fine OBU search validation-e bhalo, but test-e robust na**.

Calibration: sigmoid + isotonic

Calibration test kora hoyeche, but classification decision improvement strong na.

Best calibrated OBU model test-e:

Model	Accuracy	Balanced Acc	Macro F1	Moderate Recall	High Recall
Sigmoid calibrated OBU	0.7191	0.6417	0.5620	0.5429	0.6316
Isotonic calibrated OBU	0.7191	0.6417	0.5579	0.5429	0.6316
Baseline rethreshold	**0.7191**	**0.6641**	0.5591	0.5048	**0.7368**

Alternative Stage-2 models

Test result:

Stage-2 Model	Accuracy	Balanced Acc	Macro F1	Moderate Recall	High Recall
LR	0.7131	0.6258	0.5339	0.4952	0.6316
RF	0.7131	0.6482	0.5361	0.4571	0.7368
ExtraTrees	0.7083	0.6243	0.5157	0.4381	0.6842
HistGradientBoosting	0.7059	0.6012	0.5031	0.4476	0.6053
XGBoost	0.7143	0.6290	0.5435	0.5048	0.6316
Baseline rethreshold	**0.7191**	**0.6641**	**0.5591**	0.5048	**0.7368**

Stage-2 ensemble

Ensemble	Accuracy	Balanced Acc	Macro F1	Moderate Recall	High Recall
Mean all available	0.7143	0.6458	0.5377	0.4762	0.7105
Mean top-3 validation	0.7179	0.6385	0.5558	0.5333	0.6316
Baseline rethreshold	**0.7191**	**0.6641**	**0.5591**	0.5048	**0.7368**

Cost-sensitive Stage-2 training

Cost-sensitive training high-risk class weight baray. Kisu model high recall barate pare, but moderate class collapse kore.

Model	Accuracy	Macro F1	Moderate Recall	High Recall
LR high weight 4.0	0.6963	0.4743	0.2952	0.8158

Model	Accuracy	Macro F1	Moderate Recall	High Recall
ExtraTrees high weight 4.0	0.6771	0.4139	0.1143	0.8947
Baseline rethreshold	**0.7191**	**0.5591**	**0.5048**	0.7368

Final decision

Ei experiment theke amar decision:

Best practical model ekhono: baseline cascade + Stage-2 threshold 0.50

final selected model: baseline cascade + Stage-2 threshold 0.50 এর জন্য clean version l

notebook naam nhanes_FINAL_CLEAR_BASELINE_STAGE2_THR050_MODEL_NOTE...

ei notebook e ase

- Stage-1: **RF + RandomOverSampler gate**
- Stage-2: **OBU + TreeMedianSelector + GradientBoosting**
- Final Stage-2 threshold: **0.50**
- Old threshold **0.425 vs new threshold 0.50** comparison

Final selected model holo:

Stage-1 RF + ROS gate
→ low vs at-risk separate kore
Stage-2 OBU + TreeMedianSelector + GradientBoosting
→ at-risk group-er moddhe moderate vs high separate kore
Stage-2 final threshold = 0.50

Feature selection ki hoise

Model total 22 feature theke kaj korse.

Stage-1 top 19 feature use korse. Main important Stage-1 features:

Rank	Feature
1	age
2	blood_urea_nitrogen

Rank	Feature
3	hba1c
4	taking_bp_medicine
5	mean_sbp
6	glucose
7	hypertension_doctor_told
8	uric_acid
9	diabetes_yes
10	income_poverty_ratio

Stage-1 basically bujhte chesta kore: patient low-risk naki at-risk side-e jabe.

Stage-2 input-e top 12 feature neya hoise:

Rank	Stage-2 input feature
1	blood_urea_nitrogen
2	uric_acid
3	hemoglobin
4	serum_albumin
5	hba1c
6	mean_sbp
7	waist
8	age
9	bmi
10	mean_pulse
11	glucose
12	mean_dbp

Tarpor **TreeMedianSelector** final Stage-2 te 6ta feature select korse:

blood_urea_nitrogen
uric_acid
hemoglobin
serum_albumin
mean_sbp
age

Threshold 0.50 keno select kora hoise

Old Stage-2 threshold chilo **0.425**. New final threshold holo **0.50**.

Test set comparison:

Threshold	Accuracy	Balanced Acc	Macro F1	Moderate Recall	High Recall	High→Low Miss
Old 0.425	0.7119	0.6451	0.5322	0.4476	0.7368	2
New 0.50	**0.7191**	**0.6641**	**0.5591**	**0.5048**	**0.7368**	2

Final test performance

Final held-out test result:

Metric	Value
Accuracy	0.7191
Balanced accuracy	0.6641
Macro F1	0.5591
Weighted F1	0.7566
Low recall	0.7507
Moderate recall	0.5048
High recall	0.7368
High → Low miss count	2
Stage-1 threshold	0.4400
Stage-2 threshold	0.5000

Stage-1 gate analysis

Stage-1 gate summary:

Actual	Sent to Stage-2?	Count
High	False	2
High	True	36
Low	False	518
Low	True	172
Moderate	False	32
Moderate	True	73

Eta bujhay:

Actual high 38 jon-er moddhe Stage-1:

- 36 jon-ke at-risk bole Stage-2 te pathaise
- 2 jon-ke low gate-e rekhe dise

Ei 2 jon-i final **high→low miss**.

Actual moderate 105 jon-er moddhe:

- 73 jon Stage-2 te gese
- 32 jon low hisebe gate-e miss hoise

So moderate class improve korte hole Stage-1 gate aro sensitive korte hobe, but eta korle low false alarm aro barte pare.

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8. Error analysis

Test error decomposition:

Error group	Count
low_to_low	518
low_to_moderate	147
moderate_to_moderate	53
moderate_to_low	32

Error group	Count
high_to_high	28
low_to_high	25
moderate_to_high	20
high_to_moderate	8
high_to_low	2

Important observations:

1. **High→Low miss only 2**
   Eta bhalo. Severe error low.
2. **High→Moderate 8**
   Eta less dangerous than high→low, because at least patient at-risk category te thakche.
3. **Moderate→Low 32**
   Eta concern. Moderate class-er ekta portion low hisebe miss hocche.
4. **Low→Moderate/High false alarm 172 total**
   Low class-er 147 moderate + 25 high predict hoise. Screening system-e eta extra follow-up create korte pare.

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9. Train/validation/test comparison

Split	Accuracy	Balanced Acc	Macro F1	Low Recall	Moderate Recall	High Recall
Train	0.7126	0.6718	0.5431	0.7377	0.5377	0.7399
Validation	0.6939	0.6787	0.5300	0.7062	0.6000	0.7297
Test	0.7191	0.6641	0.5591	0.7507	0.5048	0.7368

Model ki vabe decision ney

Ei model direct 3-class classifier na. Eta cascade model.

Step 1: Stage-1 gate

Stage-1 RF + ROS model first decide kore:

Low risk naki at-risk?

At-risk mane moderate/high possibility ache.

Jodi Stage-1 bole low, final prediction low hoye jay.

Jodi Stage-1 bole at-risk, patient Stage-2 te jay.

Step 2: Stage-2 classifier

Stage-2 only at-risk area niye kaj kore.

Eta moderate vs high separate kore.

Stage-2 training-e OBU resampling use hoyeche, karon moderate/high area overlap kore and class imbalance ase.

Then TreeMedianSelector important feature select kore.

Finally GradientBoosting model moderate vs high probability estimate kore.

Step 3: Threshold decision

Stage-2 high probability jodi **0.50 or above** hoy, final prediction high.

Na hole moderate.

So final logic:

```
If Stage-1 says low:
  final = low
else:
  If Stage-2 high probability >= 0.50:
    final = high
  else:
    final = moderate
```

Then ask korlam

tahole eitar high ar low recall fixed rekhe moderate recall ar barano jabe kina?

eita kidney risk screening model hisebe performance e ar kivabe improvement ana possible

ami kivabe performance e improve korte pari

1. Current model-e moderate recall kothay loss hocche?

Moderate class loss mainly দুই জায়গায় হচ্ছে।

Problem 1: Stage-1 gate moderate-ke low kore dicche

Stage-1 summary:

Actual	Sent to Stage-2	Count
Moderate	No	32
Moderate	Yes	73

এই **32 জন moderate** Stage-1 এই low হয়ে যাচ্ছে। এরা Stage-2 তে যেতেই পারছে না। তাই Stage-2 যত improve করো, এই 32 জন recover হবে না।

So moderate recall improve করতে হলে প্রথম target হওয়া উচিত:

Stage-1 moderate rescue.

মানে Stage-1 এমনভাবে tune করতে হবে যেন actual moderate বেশি Stage-2 তে যায়, কিন্তু low recall খুব বেশি না কমে।

Problem 2: Stage-2 moderate-ke high kore dicche

Stage-2 তে যাওয়ার পর:

Error	Count
Moderate → High	20
High → Moderate	8

Stage-2 threshold 0.50 দেওয়ার পর moderate recall improve হয়েছে, কিন্তু এখনও **20 জন moderate high হয়ে যাচ্ছে।**

So second target:

Stage-2 high threshold / moderate-high boundary refine kora.

কীভাবে high/low recall fixed রেখে moderate recall বাড়ানোর চেষ্টা করা যায়?

Approach A: Constrained threshold search

এটা সবচেয়ে safe next experiment.

আগে তুমি শুধু Stage-2 threshold tune করেছিলে। এবার একসাথে দুইটা threshold tune করবে:

1. **Stage-1 threshold**
   Low vs at-risk gate.
2. **Stage-2 threshold**
   Moderate vs high decision.

Search logic:

```
For each Stage-1 threshold:
  For each Stage-2 threshold:
    calculate validation metrics

    keep only models where:
      low recall >= baseline low recall - small tolerance
      high recall >= baseline high recall - small tolerance
      high_to_low_miss <= baseline high_to_low_miss

    among valid models:
      choose highest moderate recall
```

এখানে small tolerance হিসেবে 0.00 দিলে strict fixed হবে। কিন্তু practical model selection-এ 0.01 tolerance রাখা যায়, কারণ validation/test sample ছোট।

Decision threshold tuning is a standard way to adapt classifier decisions to the target metric, especially when default thresholds are not aligned with the application objective. Scikit-learn's documentation also emphasizes choosing threshold metrics according to the use case rather than relying on default 0.5 decisions.

Approach B: Moderate rescue rule

Current model যাদের low বলছে, তাদের মধ্যে কিছু patient actual moderate। এই group recover করার জন্য একটা conservative rescue rule করা যায়।

Example logic:

```
If Stage-1 says low
but Stage-1 at-risk probability is close to threshold
and kidney-related risk signal is present:
  final prediction = moderate
```

Possible risk signal:

- BUN high pattern
- uric acid high pattern

- hemoglobin low pattern
- serum albumin low pattern
- systolic BP high pattern
- older age

তবে এই rule অবশ্যই validation set-এ tune করতে হবে। Manually rule বানিয়ে test-e optimize করা যাবে না, otherwise leakage হবে।

এটার goal হবে:

moderate_to_low কমানো
but low_to_moderate বেশি না বাড়ানো

Approach C: Three-zone Stage-1 gate

Current Stage-1 binary decision:

low OR at-risk

এটা moderate class-এর জন্য harsh। Instead, Stage-1 output probability থেকে তিনটা zone করা যায়:

Very low probability → low
Uncertain middle zone → moderate
High at-risk probability → send to Stage-2

Example:

```
If p_at_risk < t_low:
  low
elif t_low <= p_at_risk < t_high:
  moderate
else:
  Stage-2 moderate/high decision
```

এটা moderate recall বাড়াতে পারে, কারণ borderline low/at-risk cases directly moderate হবে।

But risk: low false alarm বাড়তে পারে। তাই আবার constraint লাগবে:

low recall >= 0.7507
high recall >= 0.7368

Approach D: Stage-2 ordinal/cost-sensitive model, but moderate-focused

আগের cost-sensitive experiment high-risk weight বাড়িয়ে moderate class collapse করেছিল। তাই এবার high weight না বাড়িয়ে **moderate-focused loss** try করা দরকার।

Possible class weights:

low: 1.0
moderate: 1.5 to 3.0
high: 2.0 to 3.0

কিন্তু current cascade-e Stage-2 শুধু moderate vs high। তাই Stage-2 তে moderate weight বাড়ালে moderate→high error কমতে পারে, কিন্তু high recall কমে যেতে পারে।

তাই constraint দরকার:

high recall must remain >= 0.7368

Approach E: Probability calibration + threshold, not calibration alone

তুমি calibration already try করেছো। Calibration alone final classification improve করে নাই।

কিন্তু calibration useful হতে পারে যদি তুমি এরপর constrained threshold search করো। Probability calibration is mainly for making predicted probabilities better aligned with observed frequencies, not automatically for improving class labels.

So better experiment:

Train model
Calibrate probability on validation/cross-validation
Then tune thresholds with constraints

Not:

Calibrate and directly accept default threshold

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ei experiment gulo kore ekta notebook

notebook naam nhanes_CONSTRAINED_MODERATE_RECALL_EXPERIMENTS_NOT...

এই notebook-এ আছে:

1. **Baseline reproduce**
   Current final cascade model reproduce করবে।
2. **Approach A: Constrained Stage-1 + Stage-2 threshold search**
   Low/high recall preserve রেখে moderate recall বাড়ে কিনা খুঁজবে।
3. **Approach B: Moderate rescue rule**
   Stage-1 যাদের low বলছে কিন্তু borderline risk signal আছে, তাদের conservative ভাবে moderate rescue করার experiment।

4. **Approach C: Three-zone Stage-1 gate**
   Low / uncertain moderate / at-risk এই তিন zone দিয়ে test করবে।
5. **Approach D: Moderate-focused Stage-2 cost-sensitive models**
   LR, RF, ExtraTrees, GB দিয়ে moderate-vs-high boundary improve করার চেষ্টা করবে।
6. **Approach E: Calibration + constrained threshold**
   Sigmoid/isotonic calibration করে তারপর threshold tuning করবে।
7. **Final validation-selected leaderboard + held-out test safety check**
   Validation দিয়ে candidate select করবে, test set শুধু final check হিসেবে দেখাবে।

Baseline current model

Current final model:

Stage-1 threshold = 0.44
Stage-2 threshold = 0.50

Test result:

Metric	Value
Accuracy	0.7191
Balanced accuracy	0.6641
Macro F1	0.5591
Low recall	0.7507
Moderate recall	0.5048
High recall	0.7368
High → Low miss	2

Experiment A: Joint Stage-1 + Stage-2 threshold search

Ei experiment-e Stage-1 threshold and Stage-2 threshold duita ekshathe search kora hoyeche.

Validation-e best strict candidate chilo:

Stage-1 threshold = 0.44
Stage-2 threshold = 0.50

Eta basically **same current baseline**.

Validation result:

Metric	Value
Accuracy	0.6939
Balanced accuracy	0.6787
Macro F1	0.5300
Low recall	0.7062
Moderate recall	0.6000
High recall	0.7297
High → Low miss	1

Test-e eta:

Metric	Value
Accuracy	0.7191
Balanced accuracy	0.6641
Macro F1	0.5591
Low recall	0.7507
Moderate recall	0.5048
High recall	0.7368
High → Low miss	2

Decision: Joint threshold search kono notun better threshold ber korte pare nai. Current threshold already best stable point.

Experiment B: Moderate rescue rule

Ei experiment-er idea chilo:

Stage-1 jodi low bole,
but patient borderline risk signal show kore,
tahole low theke moderate-e rescue kora.

But result-e best selected rescue config:

rescue_margin = 0.0
min_signals = 1
n_rescued_to_moderate = 0

Meaning: **rescue actually kichui kore nai.**

Tai test result baseline-er same:

Metric	Value
Accuracy	0.7191
Balanced accuracy	0.6641
Macro F1	0.5591
Low recall	0.7507
Moderate recall	0.5048
High recall	0.7368

Decision: Moderate rescue rule useful improvement dite pare nai. Jodi rescue aggressively kori, low false alarm ba high safety risk barte pare.

Experiment C: Three-zone Stage-1 gate

Ei experiment-er idea chilo:

Confident low → low
Uncertain middle zone → moderate
High at-risk → Stage-2

Validation-e eta promising chilo.

Best validation config:

t_low = 0.44
t_high = 0.54
Stage-2 threshold = 0.50

Validation result:

Metric	Baseline Validation	Three-zone Validation
Accuracy	0.6939	0.6987

Metric	Baseline Validation	Three-zone Validation
Balanced accuracy	0.6787	**0.6913**
Macro F1	0.5300	**0.5579**
Low recall	0.7062	0.7062
Moderate recall	0.6000	**0.6381**
High recall	0.7297	0.7297

Validation-e eta clearly better mone hocchilo.

But test-e:

Metric	Baseline Test	Three-zone Test
Accuracy	0.7191	**0.7203**
Balanced accuracy	**0.6641**	0.6617
Macro F1	0.5591	**0.5684**
Low recall	0.7507	0.7507
Moderate recall	0.5048	**0.5238**
High recall	**0.7368**	0.7105

Experiment D: Moderate-focused Stage-2 cost-sensitive model

Ei experiment-e moderate class weight barano hoyechilo.

Best fallback config:

ExtraTrees
moderate_weight = 1.25
high_weight = 1.00

Validation-e moderate recall onek berechilo, but high recall almost collapse kore:

Metric	Validation
Accuracy	0.6927
Balanced accuracy	0.5355
Macro F1	0.4556

Metric	Validation
Low recall	0.7062
Moderate recall	0.8190
High recall	0.0811

Test-eo same problem:

Metric	Test
Accuracy	0.7131
Balanced accuracy	0.5083
Macro F1	0.4539
Low recall	0.7507
Moderate recall	0.6952
High recall	0.0789

Experiment E: Calibration + threshold

Best fallback:

Sigmoid calibrated Stage-2
Stage-2 threshold = 0.60

Validation:

Metric	Validation
Accuracy	0.6963
Balanced accuracy	0.5683
Macro F1	0.5093
Low recall	0.7062
Moderate recall	0.8095
High recall	0.1892

Test:

Metric	Test
Accuracy	0.7143
Balanced accuracy	0.5171
Macro F1	0.4689
Low recall	0.7507
Moderate recall	0.6952
High recall	0.1053

Decision: Calibration + threshold ei form-e unsafe. Reject.

Then

4 ta experiment korsi

৪টা experiment আছে:

1. **Experiment 1: Constrained multi-objective 4-parameter threshold search**
    - t_stage1
    - t_stage2
    - high_rescue_t
    - moderate_rescue_t
    - Search A: high recall improve
    - Search B: moderate recall improve
    - Search C: all recall/macro-F1/balanced accuracy improve
2. **Experiment 2: High rescue head**
    - binary high vs non-high head
    - cascade যাদের moderate predict করে শুধু তাদের মধ্যে high rescue করবে
3. **Experiment 3: Moderate rescue head**
    - validation-locked threshold
    - uncertain high → moderate rescue
    - conservative borderline low → moderate rescue

4. **Experiment 4: Ordinal cascade**
    - Model 1: risk >= moderate
    - Model 2: risk >= high
    - decision: low / moderate / high ordered logic

notebook naam nhanes_ADVANCED_RESCUE_HEAD_ORDINAL_CASCADE_EXPERI...

Notebook output dekhe main conclusion:

Advanced experiments run successful, but কোনো new candidate strict safety goal pass করেনি। তাই final recommendation এখনও baseline cascade model রাখা

Experiment 1: 4-parameter constrained threshold search

Ei experiment **6480 threshold configuration** evaluate koreche.

Tuned parameters:

Parameter	Meaning
t_stage1	low vs at-risk gate
t_stage2	moderate vs high decision
high_rescue_t	moderate → high rescue
moderate_rescue_t	gray-zone → moderate rescue

Search A: high recall improve

Strict validation fallback candidate:

Metric	Validation	Test
Accuracy	0.6807	0.7071
Balanced accuracy	0.5908	0.5792
Macro F1	0.3972	0.4322
Low recall	0.7598	0.7884
Moderate recall	0.0667	0.1333
High recall	0.9459	0.8158

Metric	Validation	Test
High→Low miss	1	5

Eta high recall baray:

0.7368 → 0.8158

But moderate recall collapse kore:

0.5048 → 0.1333

And high→low miss:

2 → 5

So eta reject. High recall barleo severe miss count bere geche and moderate class almost destroy hoyeche.

Search B: moderate recall improve

Best selected config:

t_stage1 = 0.44
t_stage2 = 0.52
high_rescue_t = 0.35
moderate_rescue_t = 0.42

Test result:

Metric	Baseline	Search B
Accuracy	0.7191	0.7155
Balanced accuracy	0.6641	0.6546
Macro F1	0.5591	0.5516
Low recall	0.7507	0.7507
Moderate recall	0.5048	0.4762
High recall	0.7368	0.7368
High→Low miss	2	2

Low and high recall fixed thakleo moderate recall **bareni**, ulta komeche.

So Search B fail.

Search C: all recall improve

Best fallback test:

Metric	Baseline	Search C
Accuracy	0.7191	0.7359
Balanced accuracy	0.6641	0.6437
Macro F1	0.5591	0.5545
Low recall	0.7507	0.7826
Moderate recall	0.5048	0.4381
High recall	0.7368	0.7105
High→Low miss	2	5

Accuracy and low recall improve hoyeche, but moderate/high recall kome geche. High→low miss 2 theke 5 hoyeche. Screening model hisebe reject.

Experiment 1 verdict

4-parameter threshold search publishable as ablation/negative result, because eta dekhalo je clinically constrained optimization try kora hoyeche. But final model change korar moto safe improvement dey nai.

Experiment 2: High rescue head

Goal chilo:

If cascade predicts moderate
and high_head_prob is high
and Stage-2 margin small:
  moderate → high

Trained heads:

- Logistic Regression
- Random Forest
- ExtraTrees

Best selected high rescue head:

high_head_rf_balanced
head_threshold = 0.70
margin_threshold = 0.03 or 0.08

Best test result:

Metric	Baseline	High rescue head
Accuracy	0.7191	0.7179
Balanced accuracy	0.6641	0.6609
Macro F1	0.5591	0.5558
Low recall	0.7507	0.7507
Moderate recall	0.5048	0.4952
High recall	0.7368	0.7368
High→Low miss	2	2

Eta safety maintain koreche, but **high recall improve korte pare nai**.

High recall same:

0.7368 → 0.7368

Moderate recall slight down:

0.5048 → 0.4952

Experiment 2 verdict

High rescue head controlled chilo, but practically gain dey nai. Eta baseline-er almost same, slightly worse. Final model hisebe use korar reason nai.

Experiment 3: Moderate rescue head

Goal chilo:

1. uncertain high → moderate
2. borderline low → moderate, very conservative way te

Best selected config:

moderate_head_lr_balanced
head_threshold = 0.85
high_conf_upper = 0.52
near_stage1_margin = 0.0
low_to_mod_threshold = 0.70

Test result:

Metric	Baseline	Moderate rescue head
Accuracy	0.7191	0.7179
Balanced accuracy	0.6641	0.6553
Macro F1	0.5591	0.5542
Low recall	0.7507	0.7507
Moderate recall	0.5048	0.5048
High recall	0.7368	0.7105
High→Low miss	2	2

Moderate recall improve hoy nai:

0.5048 → 0.5048

High recall komeche:

0.7368 → 0.7105

Eta likely kichu true high case-ke moderate-e shift kore dise.

Experiment 3 verdict

Moderate rescue head expected benefit dey nai. Moderate recall same, high recall lower. Reject.

Experiment 4: Ordinal cascade

Eta clinically interesting chilo, karon label order ache:

low < moderate < high

Model logic:

Model 1: P(risk >= moderate)
Model 2: P(risk >= high)

If P(risk >= moderate) low → low
If P(risk >= moderate) high and P(risk >= high) low → moderate
If P(risk >= high) high → high

Best models mostly ordinal_rf_ros.

Ordinal Search A: high recall improve

Test:

Metric	Baseline	Ordinal A
Accuracy	0.7191	0.7323
Balanced accuracy	0.6641	0.6226
Macro F1	0.5591	0.5562
Low recall	0.7507	0.7768
Moderate recall	0.5048	0.4857
High recall	0.7368	0.6053
High→Low miss	2	4

Accuracy improve hoyeche, but high recall onek kome geche.

High recall: 0.7368 → 0.6053
High→Low miss: 2 → 4

Reject.

Ordinal Search B: moderate recall improve

Test:

Metric	Baseline	Ordinal B
Accuracy	0.7191	0.7263
Balanced accuracy	0.6641	0.6291
Macro F1	0.5591	0.5286
Low recall	0.7507	0.7768
Moderate recall	0.5048	0.4000
High recall	0.7368	0.7105
High→Low miss	2	4

Moderate recall ulta komeche:

0.5048 → 0.4000

Reject.

Ordinal Search C: all recall / macro-F1 improve

Test:

Metric	Baseline	Ordinal C
Accuracy	0.7191	0.7491
Balanced accuracy	0.6641	0.6105
Macro F1	0.5591	0.5554
Low recall	0.7507	0.8072
Moderate recall	0.5048	0.4190
High recall	0.7368	0.6053
High→Low miss	2	5

Eta highest accuracy dise:

0.7191 → 0.7491

But eta mainly low class better korar jonno. Minority class, especially high-risk, worse hoyeche.

Screening model-er jonno eta dangerous.

Experiment 4 verdict

Ordinal cascade conceptually strong, but current data/features/model setup-e eta **low class optimize kore**, high-risk safety maintain korte pare nai. Diagnosis/screening context-e reject.

Overall best candidate comparison

Candidate	Accuracy	Balanced Acc	Macro F1	Low Recall	Moderate Recall	High Recall	High→Low Miss	Decision
Baseline	0.7191	0.6641	0.5591	0.7507	0.5048	0.7368	2	Keep
E1 Search B	0.7155	0.6546	0.5516	0.7507	0.4762	0.7368	2	Reject
E2 High rescue	0.7179	0.6609	0.5558	0.7507	0.4952	0.7368	2	Reject
E3 Moderate rescue	0.7179	0.6553	0.5542	0.7507	0.5048	0.7105	2	Reject
E4 Ordinal A	0.7323	0.6226	0.5562	0.7768	0.4857	0.6053	4	Reject
E4 Ordinal C	**0.7491**	0.6105	0.5554	**0.8072**	0.4190	0.6053	5	Reject

Final interpretation

Tomar 4 experiment-er answer:

Experiment	Result
4-parameter threshold search	Interesting but no strict safe improvement
High rescue head	High recall improve kore nai
Moderate rescue head	Moderate improve kore nai, high recall komse
Ordinal cascade	Accuracy barse, but high-risk safety worse

Therefore:

Current baseline cascade is still the safest final model.

Final model:

Stage-1 threshold = 0.44
Stage-2 threshold = 0.50

Final test performance:

Accuracy = 0.7191
Balanced accuracy = 0.6641
Macro F1 = 0.5591
Low recall = 0.7507
Moderate recall = 0.5048
High recall = 0.7368
High→Low miss = 2
